



Cadillac Utilities Department
200 Lake St.
Cadillac, MI 49601
231-775-0181
231-775-8755

WATER RESOURCES
DIVISION
MAR 26 2014

CADILLAC DISTRICT

Mr. Brian Jankowski
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac, Michigan 49601

Re: 2013'S POLLUTANT MINIMIZATION PLAN YEARLY STATUS REPORT

Dear Mr. Jankowski,

Enclosed, please find the Cadillac WTP 2013 Pollutant Minimization Plan Yearly Status Report. I have enclosed the Influent and Effluent Mercury results for 2011-2013, also included is a report of mercury levels in Cadillac's Bio-solids.

There was no industry that came to town that had a potential of new mercury source and all of the previous potential mercury sources have not changed their processes. Because of this there is no updated list of potential mercury sources for the City of Cadillac.

A close Examination of the effluent data will reveal an average mercury concentration below the 0.583 ng/l for 2011-2013 and an average concentration below 0.60 ng/l for 2013. This level of mercury meets the goal of the Pollutant Minimization Program. We will continue to monitor the level of mercury in our effluent per our NPDES permit.

Please feel free to call if you have any questions.

Sincerely,

Hamid Rahbarnoohi, Water Resources Supervisor
Cadillac WTP

Phone #: (231) 775-2841
Fax #: (231) 775-2876

E-mail: hrahbarnoohi@cadillac-mi.net
Cc: File -Mercury Minimization reports

RECEIVED MAR 26 2014

List of Potential Mercury Sources **2013**

1. AAR Factory	AAR 201 Haynes Street
2. AAR Rinse Tank	AAR 201 Haynes Street
3. AAR 6 th	AAR 6 th Street
4. Avon Automotive	Avon 603 West 7 th Street
5. Borg Warner	Borg Warner 1100 Wright Street
6. Paulstra	Paulstra 600 Seventh Street
7. Michigan Rubber	Michigan Rubber 1200 Eighth Street
8. Rec Boat Holdings*	Rec Boat Holdings 609 West 13 th Street
9. Rec Boat Holding Sport*	Rec Boat Holdings Sport 925 Frisbie Street
10. Rec Boat Holdings Trailer*	Rec Boat Holdings Trailer 1500 Ron Wilson
11. Cadillac Castings Inc.	CCI 1500 Fourth Street
12. Wexford County Landfill, LLC**	900 N. US HWY 131, Manton, MI 49663
13. Cadillac Renewable Energy	CRE, 1525 Frank Millner
14. Mercy Hospital	Mercy Hospital 400 Hobart Street
15. FIAMM	FIAMM Technologies 946 Frisbie Street
16. ARVCO	ARVCO 1355 Marty Paul Street
17. Piranha Hose	Piranha Hose 2500 Raymond A. Wiegler St.
18. Avon Protection	Avon Protection 503 8 th Street

*Rec Boat Holdings formerly known as 4-Winns

**Wexford County Landfill, LLC is owned currently by American Waste, formerly owned by Wexford County

City of Cadillac

Mercury, Influent and Effluent concentration through Cadillac Wastewater Treatment Plant (2011-2013)

Date	Effluent, Hg, ng/L, EPA Method 1631	Influent Hg, mg/L, Low level Hg
11/4/13	<0.5	0.0007
8/12/13	<0.5	
8/1/13		<0.0002
5/6/13	<0.55	<0.0002
2/6/13	<.85	<0.0002
11/5/12	<0.5	<0.0002
8/1/12	<0.5	
5/9/12	0.85	
2/2/12	0.65	
11/1/11	<0.5	<0.0005
8/10/11	<0.5	<0.0002
5/2/11	<0.55	<0.0002
2/7/11	<0.55	<0.0002

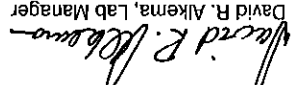
Mercury Concentration in Biosolids, 2011-2013

Sample Date	Mercury Conc. mg of Hg/Kg total solids
10/17/13	0.74
4/15/13	0.53
9/10/12	0.94
4/02/12	0.90
9/14/11	1.22
5/5/11	1.66

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No.: 138595
Date Reported: 11/01/13

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample for Fall Biosolids.

Sample ID: "Digester #4"

Sampled By: DV of City of Cadillac, Water Resources

Sample Date: 10/17/13

Sample Time: 1400

KAR Sample No.: 138595-01

Sample Type: sludge

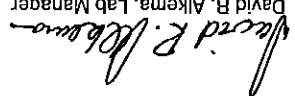
Date Received: 10/18/13

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	10/28/13	ALK	
Prep, metals	Completed		EPA 3050B	10/24/13	BPK	
Aluminum, total	4870	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 200
Arsenic, total	36	mg/kg total solids	EPA 200.8	11/01/13	ZZW	Sample is not homogeneous; results approximate. Detection limit = 5
Cadmium, total	<10	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 200
Calcium, total	30,300	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 200
Chromium, total	79.6	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 20
Copper, total	1080	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 40
Lead, total, by ICP	<100	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 40
Magnesium, total	3340	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 200
Mercury, total	0.74	mg/kg total solids	EPA 7471A	10/28/13	ALK	Detection limit = 0.05
Molybdenum, total	<40	mg/kg total solids	EPA 200.7	11/01/13	CLB	
Nickel, total	<40	mg/kg total solids	EPA 200.7	11/01/13	CLB	
Potassium, total	3250	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 200
Selenium, total	10	mg/kg total solids	EPA 200.8	11/01/13	ZZW	Sample is not homogeneous; results approximate. Detection limit = 5
Silver, total	<0.05	mg/kg total solids	EPA 6010B	10/30/13	BPK	
Sodium, total	6190	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 2000
Zinc, total	1430	mg/kg total solids	EPA 200.7	11/01/13	CLB	Detection limit = 40
Chloride	8200	mg/kg total solids	EPA 300.0A	10/28/13	JHB	Detection limit = 180
Nitrogen, ammonia	35,100	mg/kg total solids	EPA 350.1	10/30/13	JWW	Detection limit = 690
Nitrogen, nitrate	<35	mg/kg total solids	EPA 353.2	10/25/13	ALK	
Nitrogen, total kjeldahl	72,800	mg/kg total solids	EPA 351.2	10/25/13	ALK	Detection limit = 3500
Phosphorus, total (as P)	45,600	mg/kg total solids	SM 4500-P B5.E	10/23/13	JHB	Detection limit = 3500
Sulfate	546	mg/kg total solids	EPA 300.0A	10/28/13	JHB	Detection limit = 350
Waste pH	7.8	S.U.	EPA 9045C	10/21/13	AMG	
Solids, total	2.90	% by weight	EPA 160.3	10/21/13	AMG	Detection limit = 0.01
Solids, volatile	48.03	% of total solids	EPA 160.4	10/21/13	AMG	Detection limit = 0.01

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No.: 132637
Date Reported: 04/30/13

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one Spring Biosolids sample.

Sample ID: "Digester #4"

Sampled By: DC of City of Cadillac, Water Resources

Sample Date: 04/15/13

Sample Time: 1400

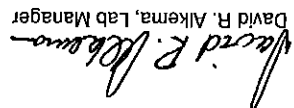
Date Received: 04/16/13
Sample Type: sludge
KAR Sample No.: 132637-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/19/13	DBL	
Prep, metals	Completed		EPA 3050B	04/19/13	DBL	
Aluminum, total	4130	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 59
Arsenic, total	36.9	mg/kg total solids	EPA 6020	04/19/13	PML	Detection limit = 3
Cadmium, total	2.7	mg/kg total solids	EPA 6020	04/19/13	PML	Detection limit = 0.6
Calcium, total	30,800	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 118
Chromium, total	92	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 12
Copper, total	950	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 24
Lead, total, by ICP	<59	mg/kg total solids	EPA 6010B	04/19/13	DBL	
Magnesium, total	4500	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 118
Mercury, total	0.53	mg/kg total solids	EPA 7471A	04/19/13	DBL	Detection limit = 0.10
Molybdenum, total	<24	mg/kg total solids	EPA 6010B	04/19/13	DBL	
Nickel, total	<24	mg/kg total solids	EPA 6010B	04/19/13	DBL	
Potassium, total	4040	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 118
Selenium, total	3.6	mg/kg total solids	EPA 6020	04/19/13	PML	Detection limit = 3
Silver, total	<6	mg/kg total solids	EPA 6010B	04/19/13	DBL	
Sodium, total	7850	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 1180
Zinc, total	1130	mg/kg total solids	EPA 6010B	04/19/13	DBL	Detection limit = 24
TCLP extraction	Completed		EPA 1311	04/18/13	JWW	
TCLP report	See comment			04/30/13	GJE	This material does not exhibit the Toxicity Characteristic with respect to the requested TCL parameters.
ZHE extraction	Completed		EPA 1311	04/18/13	JWW	
Chloride	9500	mg/kg total solids	EPA 300.0A	04/23/13	ALK	Detection limit = 1100
Nitrogen, ammonia	37,300	mg/kg total solids	EPA 350.1	04/24/13	JWW	Detection limit = 820
Nitrogen, nitrate	<41	mg/kg total solids	EPA 353.2	04/18/13	ALK	
Nitrogen, total kjeldahl	79,100	mg/kg total solids	EPA 351.2	04/22/13	JAR	Detection limit = 2100
Phosphorus, total (as P)	29,000	mg/kg total solids	SM 4500-P B5.E	04/18/13	JHB	Detection limit = 820
Sulfate	506	mg/kg total solids	EPA 300.0A	04/23/13	ALK	Detection limit = 410
Waste pH	7.6	S.U.	EPA 9045C	04/17/13	JWW	
Soilids, total	2.44	% by weight	EPA 160.3	04/17/13	JWW	Detection limit = 0.01
Solids, volatile	58.28	% of total solids	EPA 160.4	04/17/13	JWW	Detection limit = 0.01
Prior, Poll, volatiles	See below		EPA 8260	04/22/13	JAR	
Prep, VOA	Completed		EPA 5030	04/22/13	JAR	
1,1,1-Trichloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,1,2,2-Tetrachloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,1,2-Trichloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,1-Dichloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,1-Dichloroethene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,2-Dichlorobenzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,2-Dichloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,2-Dichloropropane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,3-Dichlorobenzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
1,4-Dichlorobenzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 132637

Attest: 
David R. Alkema, Lab Manager

Project Description: Analysis of one Spring Biosolids sample.

Sample ID : "Digester #4"

Sampled By : DC of City of Cadillac, Water Resources

Date Received : 04/16/13

Sample Type : sludge

Sample Time : 1400

KAR Sample No. : 132637-01

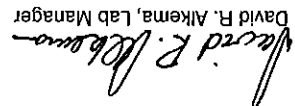
Test	Result	Units of Measure	Method	Analyzed	Analysis	Comments
2-Chloroethylvinyl ether	<10,000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Acrolein	<20,000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Acrylonitrile	<20,000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Benzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Bromodichloromethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Bromoform	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Bromomethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Carbon tetrachloride	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Chlorobenzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Chloroethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Chloroform	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Chloromethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Cis-1,3-Dichloropropene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Dibromochloromethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Ethylbenzene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Methylene chloride	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Tetrachloroethene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Toluene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Trans-1,2-Dichloroethene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Trans-1,3-Dichloropropene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Trichloroethene	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Trichlorofluoromethane	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Vinyl chloride	<5000	ug/kg total solids	EPA 8260	04/22/13	JAR	
Prior. Pol. acids	See below		EPA 8270	04/23/13	GMB	
Prior. Pol. base-neutrals	See below		EPA 8270	04/23/13	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	04/19/13	GMB	
1,2,4-Trichlorobenzene 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
1,2-Dichlorobenzene by 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
1,2-Diphenylhydrazine	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
1,3-Dichlorobenzene by 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
1,4-Dichlorobenzene by 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,3,7,8-TCDD by 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,4,6-Trichlorophenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,4-Dichlorophenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,4-Dimethylphenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,4-Dinitrophenol	<50,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,4-Dinitrotoluene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2,6-Dinitrotoluene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2-Chloronaphthalene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2-Chlorophenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2-Methyl-4,6-dinitrophenol	<50,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
2-Nitrophenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
3,3'-Dichlorobenzidine	<50,000	ug/kg total solids	EPA 8270	04/23/13	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 132637
Date Reported : 04/30/13

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one Spring Biosolids sample.

Sample ID : "Digester #4"

Sampled By : DC of City of Cadillac, Water Resources

Sample Date : 04/15/13

Sample Time : 1400

Date Received : 04/16/13
Sample Type : sludge
KAR Sample No. : 132637-01

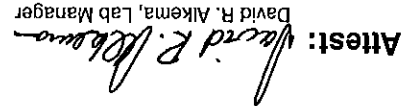
Test	Result	Units of Measure	Method	Analysed	Analysed	Comments
4-Bromophenyl phenyl ether	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
4-Chloro-3-methylphenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
4-Chlorophenyl phenyl ether	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
4-Nitrophenol	<50,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Acenaphthylene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Acenaphthene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Anthracene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzidine	<125,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzo(a)anthracene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzo(a)pyrene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzo(b)fluoranthene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzo(g,h,i)perylene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Benzo(k)fluoranthene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Bis(2-chloroethoxy)methane	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Bis(2-chloroethyl)ether	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Bis(2-chloroisopropyl)ether	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Bis(2-ethylhexyl)phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Butylbenzyl phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Chrysene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Dt-n-butyl phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Dt-n-octyl phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Dibenzo(a,h)anthracene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Diethyl phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Dimethyl phthalate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Fluoranthene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Fluorene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Hexachlorobenzene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Hexachlorobutadiene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Hexachlorocyclopentadiene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Hexachloroethane	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Indeno(1,2,3cd)pyrene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Isophorone	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
N-Nitrosodipropylamine	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
N-Nitrosodimethylamine	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
N-Nitrosodiphenylamine	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Naphthalene by Method 8270	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Nitrobenzene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Pentachlorophenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Phenanthrene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Phenol	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Pyrene	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Prior. Poll. PCB's	See below		EPA 8270	04/23/13	GMB	
4,4'-DDD	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 132637
Date Reported : 04/30/13

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one Spring Biosolids sample.

Sample ID : "Digester #4"

Sampled By : DC of City of Cadillac, Water Resources

Sample Date : 04/15/13

Sample Time : 1400

Date Received : 04/16/13
Sample Type : sludge
KAR Sample No. : 132637-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4,4'-DDE	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
4,4'-DDT	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Aldrin	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Alpha-BHC	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Beta-BHC	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Chlordane	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Delta-BHC	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Dieldrin	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Endosulfan I	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Endosulfan II	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Endosulfan sulfate	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Endrin	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Endrin aldehyde	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Gamma-BHC	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Heptachlor	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Heptachlor epoxide	<25,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1016 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1221 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1232 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1242 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1248 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1254 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
PCB Aroclor 1260 by 8270	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	
Toxaphene	<250,000	ug/kg total solids	EPA 8270	04/23/13	GMB	

Sample ID : TCLP Leachate of Digester #4

Sampled By :
Sample Date :
Sample Time :
Date Received : 04/16/13
Sample Type : TCLP
KAR Sample No. : 132637-01T

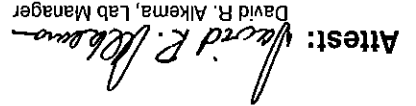
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	See below		EPA 6010B	04/22/13	DBL	
Prep. Hg	Completed		EPA 7470A	04/23/13	DBL	
Prep. metals	Completed		EPA 200.7/8	04/22/13	DBL	
Arsenic, total, by ICP	<0.5	mg/L	EPA 6010B	04/22/13	DBL	TC regulatory limit is 5.0 mg/L
Barium, total	0.30	mg/L	EPA 6010B	04/22/13	DBL	TC regulatory limit is 100 mg/L
Cadmium, total	<0.025	mg/L	EPA 6010B	04/22/13	DBL	TC regulatory limit is 1.0 mg/L
Chromium, total	<0.05	mg/L	EPA 6010B	04/22/13	DBL	TC regulatory limit is 5.0 mg/L
Copper, total	<0.1	mg/L	EPA 6010B	04/22/13	DBL	
Lead, total, by ICP	<0.25	mg/L	EPA 6010B	04/22/13	DBL	TC regulatory limit is 5.0 mg/L

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No.: 132637
Date Reported: 04/30/13

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one Spring Biosolids sample.

Sample ID: ZHE Leachate of Digester #4

Sampled By: Date Received: 04/16/13
Sample Date: Sample Type: ZHE
Sample Time: KAR Sample No.: 132637-01Z

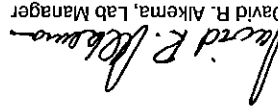
Test	Result	Units of Measure	Method	Analyzed	Analysis	Comments
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	04/22/13	JAR	TC regulatory limit is 0.2 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No.: 126295
Date Reported: 09/25/12

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample for Fall Biosolids.

Sample ID: "Digester"

Sampled By: ES of City of Cadillac, Water Resources

Sample Date: 09/10/12

Sample Time: 0900

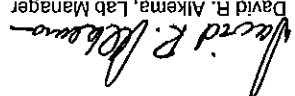
Date Received: 09/11/12
Sample Type: sludge
KAR Sample No.: 126295-01

Test	Result	Units of Measure	Method	Analyzed	Analysis	Comments
Prep. Hg	Completed		EPA 7471A	09/25/12	DBL	
Prep. metals	Completed		EPA 3050	09/20/12	PML	
Aluminum, total	5840	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 51
Arsenic, total	28.6	mg/kg total solids	EPA 6020	09/21/12	PML	Detection limit = 1.5
Cadmium, total	3.1	mg/kg total solids	EPA 6020	09/21/12	PML	Detection limit = 0.3
Calcium, total	28,700	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 51
Chromium, total	105	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 5
Copper, total	1040	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 10
Lead, total, by ICP	33	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 25
Magnesium, total	4180	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 51
Mercury, total	0.94	mg/kg total solids	EPA 7471A	09/25/12	DBL	Detection limit = 0.08
Molybdenum, total	45	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 10
Nickel, total	25	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 10
Potassium, total	2340	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 51
Selenium, total	5.2	mg/kg total solids	EPA 6020	09/21/12	PML	Detection limit = 1.5
Silver, total	5.6	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 2.6
Sodium, total	4130	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 510
Zinc, total	1250	mg/kg total solids	EPA 6010B	09/21/12	DBL	Detection limit = 10
Chloride	5820	mg/kg total solids	EPA 300.0A	09/21/12	ALK	Detection limit = 130
Nitrogen, ammonia	41,900	mg/kg total solids	EPA 350.1	09/13/12	JWW	Detection limit = 5200
Nitrogen, nitrate	<26	mg/kg total solids	EPA 353.2	09/12/12	ALK	
Nitrogen, total kjeldahl	73,700	mg/kg total solids	EPA 351.2	09/17/12	JAR	Detection limit = 2600
Phosphorus, total (as P)	46,000	mg/kg total solids	SM 4500-P B5.E	09/18/12	JHB	Detection limit = 2600
Sulfate	6930	mg/kg total solids	EPA 300.0A	09/21/12	ALK	Detection limit = 260
Waste pH	8.2	S.U.	EPA 9045C	09/25/12	HES	
Solids, total	3.90	% by weight	EPA 160.3	09/12/12	JWW	Detection limit = 0.01
Solids, volatile	55.63	% of total solids	EPA 160.4	09/12/12	JWW	Detection limit = 0.01

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No.: 121966
Date Reported: 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample.

Sample ID: "Digester #4"

Sampled By: DG of City of Cadillac, Water Resources

Sample Date: 04/02/12

Sample Time: 1330

KAR Sample No.: 121966-01
Sample Type: sludge
Date Received: 04/03/12

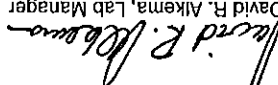
Test	Result	Units of Measure	Method	Analyzed	Analysis	Comments
Prep, Hg	Completed		EPA 7471A	04/06/12	DBL	
Prep, metals	Completed		EPA 30xx,200.x	04/10/12	DBL	
Aluminum, total	5650	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 120
Arsenic, total	30.0	mg/kg total solids	EPA 6020	04/13/12	PML	Detection limit = 6
Cadmium, total	4.2	mg/kg total solids	EPA 6020	04/13/12	PML	Detection limit = 1.5
Calcium, total	39,300	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 240
Chromium, total	106	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 24
Copper, total	1020	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 48
Lead, total, by ICP	<60	mg/kg total solids	EPA 6010B	04/11/12	DBL	
Magnesium, total	5220	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 240
Mercury, total	0.90	mg/kg total solids	EPA 7471A	04/06/12	DBL	Detection limit = 0.10
Molybdenum, total	61	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 48
Nickel, total	<48	mg/kg total solids	EPA 6010B	04/11/12	DBL	
Potassium, total	4920	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 240
Selenium, total	8.3	mg/kg total solids	EPA 6020	04/13/12	PML	Detection limit = 6
Silver, total	<12	mg/kg total solids	EPA 6010B	04/11/12	DBL	
Sodium, total	8750	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 2400
Zinc, total	1560	mg/kg total solids	EPA 6010B	04/11/12	DBL	Detection limit = 48
TCLP extraction	Completed		EPA 1311	04/09/12	JWW	
TCLP report	See comment			04/17/12	SRM	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZnE extraction	Completed		EPA 1311	04/09/12	JWW	
Chloride	13,100	mg/kg total solids	EPA 300.0A	04/10/12	ALK	Detection limit = 280
Nitrogen, ammonia	50,000	mg/kg total solids	EPA 350.1	04/12/12	JWW	Detection limit = 1100
Nitrogen, nitrate	<5.5	mg/kg total solids	EPA 353.2	04/05/12	ALK	
Nitrogen, total kjeldahl	93,800	mg/kg total solids	EPA 351.2	04/09/12	JAR	Detection limit = 2700
Phosphorus, total (as P)	38,100	mg/kg total solids	SM 4500-P B5.E	04/10/12	JHB	Detection limit = 1100
Sulfate	20,100	mg/kg total solids	EPA 300.0A	04/10/12	ALK	Detection limit = 2800
Waste pH	8.0	S.U.	EPA 9045C	04/11/12	GJE	
Solids, total	1.85	% by weight	EPA 160.3	04/05/12	JWW	Detection limit = 0.01
Solids, volatile	59.46	% of total solids	EPA 160.4	04/05/12	JWW	Detection limit = 0.01
Prior, Pol, volatiles	See below		EPA 8260	04/11/12	KTL	
Prep, VOA	Completed		EPA 5030	04/11/12	KTL	
1,1,1-Trichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,1,2-Trichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,1,2-Trichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,1-Dichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,2-Dichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,2-Dichloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,2-Dichloropropane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,3-Dichlorobenzene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
1,4-Dichlorobenzene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 121966
Date Reported : 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project Description: Analysis of one sludge sample.

Sample ID : "Digester #4"

Sampled By : DG of City of Cadillac, Water Resources

Sample Date : 04/02/12

Sample Time : 1330

Date Received : 04/03/12
Sample Type : sludge
KAR Sample No. : 121966-01

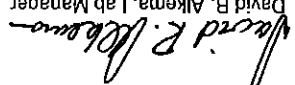
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Chloroethylvinyl ether	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Acrolein	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Acrylonitrile	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Benzene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Bromodichloromethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Bromoform	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Bromomethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Carbon tetrachloride	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Chlorobenzene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Chloroethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Chloroform	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Chloromethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Cis-1,3-Dichloropropene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Dibromochloromethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Ethylbenzene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Methylene chloride	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Tetrachloroethene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Toluene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Trans-1,2-Dichloroethene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Trans-1,3-Dichloropropene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Trichloroethene	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Trichlorofluoromethane	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Vinyl chloride	<5000	ug/kg total solids	EPA 8260	04/11/12	KTL	
Prior. Poll. acids	See below		EPA 8270	04/13/12	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	04/13/12	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	04/04/12	GMB	
1,2,4-Trichlorobenzene 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
1,2-Dichlorobenzene by 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
1,2-Diphenylhydrazine	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
1,3-Dichlorobenzene by 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
1,4-Dichlorobenzene by 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,3,7,8-TCDD by 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,4,6-Trichlorophenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,4-Dimethylphenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,4-Dinitrophenol	<60,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,4-Dinitrotoluene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2,6-Dinitrotoluene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2-Chloronaphthalene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2-Chlorophenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2-Methyl-4,6-dinitrophenol	<60,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
2-Nitrophenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
3,3'-Dichlorobenzidine	<60,000	ug/kg total solids	EPA 8270	04/13/12	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 121966
Date Reported : 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project Description: Analysis of one sludge sample.

Sample ID : "Digester #4"

Sampled By : DG of City of Cadillac, Water Resources

Sample Date : 04/02/12

Sample Time : 1330

Date Received : 04/03/12
Sample Type : sludge
KAR Sample No. : 121966-01

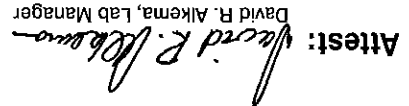
Test	Result	Units of Measure	Method	Analyzed	Analysis	Comments
4-Bromophenyl phenyl ether	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
4-Chloro-3-methylphenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
4-Chlorophenyl phenyl ether	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
4-Nitrophenol	<60,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Acenaphthene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Acenaphthylene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Anthracene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzidine	<150,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzo(a)anthracene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzo(a)pyrene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzo(b)fluoranthene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzo(g,h,i)perylene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Benzo(k)fluoranthene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Bis(2-chloroethoxy)methane	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Bis(2-chloroethyl)ether	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Bis(2-chloroisopropyl)ether	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Bis(2-ethylhexyl)phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Butylbenzyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Chrysene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
D-n-butyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
D-n-octyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Diethyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Diethyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Dimethyl phthalate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Fluoranthene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Fluorene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Hexachlorobenzene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Hexachlorobutadiene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Hexachlorocyclopentadiene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Hexachloroethane	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Indeno(1,2,3-cd)pyrene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Isophorone	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
N-Nitrosod-n-propylamine	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
N-Nitrosodimethylamine	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
N-Nitrosodiphenylamine	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Naphthalene by Method 8270	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Nitrobenzene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Penachlorophenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Phenanthrene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Phenol	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Pyrene	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Prior. Poll. pest./PCBs	See below		EPA 8270	04/13/12	GMB	
4,4'-DDD	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 121966
Date Reported : 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project Description: Analysis of one sludge sample.

Sample ID : "Digester #4"

Sampled By : DG of City of Cadillac, Water Resources

Sample Date : 04/02/12

Sample Time : 1330

Date Received : 04/03/12
Sample Type : sludge
KAR Sample No. : 121966-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4,4'-DDE	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
4,4'-DDT	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Aldrin	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Alpha-BHC	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Beta-BHC	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Chlordane	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Delta-BHC	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Dieldrin	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Endosulfan I	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Endosulfan II	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Endosulfan sulfate	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Endrin	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Endrin aldehyde	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Gamma-BHC	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Heptachlor	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Heptachlor epoxide	<30,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1016 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1221 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1232 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1242 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1248 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1254 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
PCB Aroclor 1260 by 8270	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	
Toxaphene	<300,000	ug/kg total solids	EPA 8270	04/13/12	GMB	

Sample ID : TCLP Leachate of Digester #4

Sampled By :
Sample Date :
Sample Time :
Date Received : 04/03/12
Sample Type : TCLP
KAR Sample No. : 121966-01T

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	See below		EPA 6010B	04/11/12	DBL	
Prep. Hg	Completed		EPA 7470A	04/11/12	DBL	
Prep. metals	Completed		EPA 30xx.200.x	04/11/12	DBL	
Arsenic, total, by ICP	<0.5	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 5.0 mg/L.
Barium, total	0.53	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 1.0 mg/L.
Copper, total	<0.1	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 5.0 mg/L.
Lead, total, by ICP	<0.25	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 5.0 mg/L.

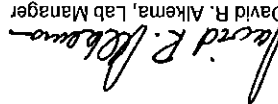
KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 121966

Date Reported : 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample.

TCLP Leachate of Digester #4

Sample ID :
Sampled By :
Sample Date :
Sample Time :
Date Received : 04/03/12
Sample Type : TCLP
KAR Sample No. : 121966-01T

Test	Result	Units of Measure	Method	Analyzed	Analys	Comments
Mercury, total	<0.05	mg/L	EPA 7470A	04/11/12	DBL	TC regulatory limit is 0.2 mg/L
Selenium, total, by ICP	<0.5	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 1.0 mg/L
Silver, total	<0.025	mg/L	EPA 6010B	04/11/12	DBL	TC regulatory limit is 5.0 mg/L
Zinc, total	0.66	mg/L	EPA 6010B	04/11/12	DBL	
TC Semi-Volatiles	See below		EPA 8270	04/13/12	GMB	
Prep, SV Acid/BN	Completed		EPA 3510	04/10/12	GMB	
Prep, herbicides	Completed		EPA 8151	04/10/12	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 400 mg/L
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 2.0 mg/L
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 0.13 mg/L
Cresols	<0.2	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 200 mg/L
Hexachlorobenzene	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 0.13 mg/L
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 0.5 mg/L
Hexachloroethane	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 3.0 mg/L
Nitrobenzene	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 2.0 mg/L
Pentachlorophenol	<0.1	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 100 mg/L
Pyridine	<0.2	mg/L	EPA 8270	04/13/12	GMB	TC regulatory limit is 5.0 mg/L
TC Pesticides by ECD	See below		EPA 8081	04/11/12	GMB	
Prep, ECD	Completed		EPA 3510	04/10/12	GMB	
Chlordane	<0.02	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.03 mg/L
Endrin	<0.004	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.02 mg/L
Hepachlor	<0.004	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.008 mg/L
Hepachlor epoxide	<0.004	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.008 mg/L
Lindane by ECD	<0.004	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.4 mg/L
Methoxychlor	<0.02	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 10.0 mg/L
Toxaphene	<0.04	mg/L	EPA 8081	04/11/12	GMB	TC regulatory limit is 0.5 mg/L
TC Herbicides	See below		EPA 8151	04/11/12	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	04/11/12	GMB	TC regulatory limit is 1.0 mg/L
2,4-D	<0.2	mg/L	EPA 8151	04/11/12	GMB	TC regulatory limit is 10.0 mg/L

ZHE Leachate of Digester #4

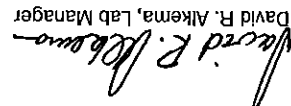
Sample ID :
Sampled By :
Sample Date :
Sample Time :
Date Received : 04/03/12
Sample Type : ZHE
KAR Sample No. : 121966-01Z

Test	Result	Units of Measure	Method	Analyzed	Analys	Comments
TC Volatiles	See below		EPA 8260	04/11/12	KTL	
Prep, VOA	Completed		EPA 5030	04/11/12	KTL	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.7 mg/L
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.5 mg/L

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 121966
Date Reported : 04/17/12

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample.

ZHE Leachate of Digester #4

Sample ID :
Sampled By :
Sample Date :
Sample Time :
Date Received : 04/03/12
Sample Type : ZHE
KAR Sample No. : 121966-01Z

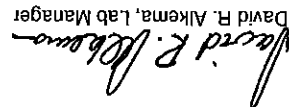
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 7.5 mg/L
Benzene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.5 mg/L
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.5 mg/L
Chlorobenzene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 100 mg/L
Chloroform	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 6.0 mg/L
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 200 mg/L
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.7 mg/L
Trichloroethene	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.5 mg/L
Vinyl chloride	<0.1	mg/L	EPA 8260	04/11/12	KTL	TC regulatory limit is 0.2 mg/L

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 113960
Date Reported : 09/29/11

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample for Fall Biosolids.

Sample ID : "Digester"

Sampled By : DR of City of Cadillac, Water Resources

Date Received : 09/15/11
Sample Type : sludge
KAR Sample No. : 113960-01

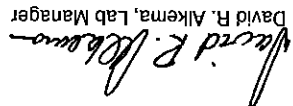
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	09/22/11	DBL	
Prep, metals	Completed		EPA 3050	09/26/11	PML	
Aluminum, total	6010	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 115
Arsenic, total	29.3	mg/kg total solids	EPA 6020	09/29/11	PML	Detection limit = 3
Cadmium, total	5.9	mg/kg total solids	EPA 6020	09/29/11	PML	Detection limit = 0.6
Calcium, total	36,800	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 115
Chromium, total	88.9	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Copper, total	946	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Lead, total, by ICP	40.4	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Magnesium, total	4430	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 115
Mercury, total	1.22	mg/kg total solids	EPA 7471A	09/22/11	DBL	Detection limit = 0.16
Molybdenum, total	53.2	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Nickel, total	23.7	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Potassium, total	3230	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 115
Selenium, total	4.8	mg/kg total solids	EPA 6020	09/29/11	PML	Detection limit = 3
Silver, total	7.6	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 6
Sodium, total	4420	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 115
Zinc, total	1510	mg/kg total solids	EPA 6010B	09/28/11	PML	Detection limit = 12
Chloride	5730	mg/kg total solids	EPA 300.0A	09/26/11	ALK	Detection limit = 147
Nitrogen, ammonia	29,800	mg/kg total solids	EPA 350.1	09/23/11	JWW	Detection limit = 585
Nitrogen, nitrate	<30	mg/kg total solids	EPA 353.2	09/27/11	ALK	
Nitrogen, total kjeldahl	69,800	mg/kg total solids	EPA 351.2	09/26/11	JHB	Detection limit = 2930
Phosphorus, total (as P)	35,100	mg/kg total solids	SM 4500-P B5,E	09/23/11	JHB	Detection limit = 2930
Sulfate	350	mg/kg total solids	EPA 9045C	09/26/11	ALK	Detection limit = 300
Waste pH	7.9	S.U.	EPA 9045C	09/26/11	HES	
Solids, total	3.42	% by weight	EPA 160.3	09/16/11	JWW	Detection limit = 0.01
Solids, volatile	57.09	% of total solids	EPA 160.4	09/16/11	JWW	Detection limit = 0.01

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: City of Cadillac, Water Resources

KAR Project No. : 111792
Date Reported : 05/10/11

Attest: 
David R. Alkema, Lab Manager

Project
Description: Analysis of one sludge sample for Spring Biosolids.

Sample ID : "Sig-1"

Sampled By : HR of City of Cadillac, Water Resources

Sample Date : 05/05/11

Sample Time : 1341

Date Received : 05/06/11
Sample Type : sludge
KAR Sample No. : 111792-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	05/10/11	PML	
Prep, metals	Completed		EPA 3050	05/06/11	PML	
Aluminum, total	2900	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Arsenic, total	30.1	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Cadmium, total	8.7	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 0.6
Calcium, total	16,500	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Chromium, total	108	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Copper, total	976	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 30
Lead, total, by ICP	35.4	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Magnesium, total	2290	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Mercury, total	1.66	mg/kg total solids	EPA 7471A	05/10/11	PML	Detection limit = 0.1
Molybdenum, total	43.0	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Nickel, total	25.3	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Potassium, total	1800	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Selenium, total	6.0	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Silver, total	7.2	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Sodium, total	2900	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Zinc, total	1710	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 30
Chloride	6700	mg/kg total solids	EPA 300.0A	05/06/11	ALK	Detection limit = 138
Nitrogen, ammonia	28,700	mg/kg total solids	EPA 350.1	05/10/11	ALK	Detection limit = 689
Nitrogen, nitrate	<28	mg/kg total solids	EPA 353.2	05/06/11	ALK	
Nitrogen, total kjeldahl	80,900	mg/kg total solids	EPA 351.2	05/06/11	JHB	Detection limit = 2760
Phosphorus, total (as P)	37,800	mg/kg total solids	SM 4500-P B5.E	05/06/11	JHB	Detection limit = 1380
Sulfate	1920	mg/kg total solids	EPA 300.0A	05/06/11	ALK	Detection limit = 276
Waste pH	7.9	S.U.	EPA 9045C	05/06/11	HES	
Solids, total	3.63	% by weight	EPA 160.3	05/06/11	JHB	Detection limit = 0.01
Solids, volatile	59.55	% of total solids	EPA 160.4	05/06/11	JHB	Detection limit = 0.01

KAR Laboratories, Inc.